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Growth and stem cells

Today I can… ☐ I can distinguish making more cells from cells taking on a specialised job. ☐ I can compare animal growth with plant growth. ☐ I can identify a role for an animal stem cell or a plant meristem.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

Lesson strips

Growth and stem cells

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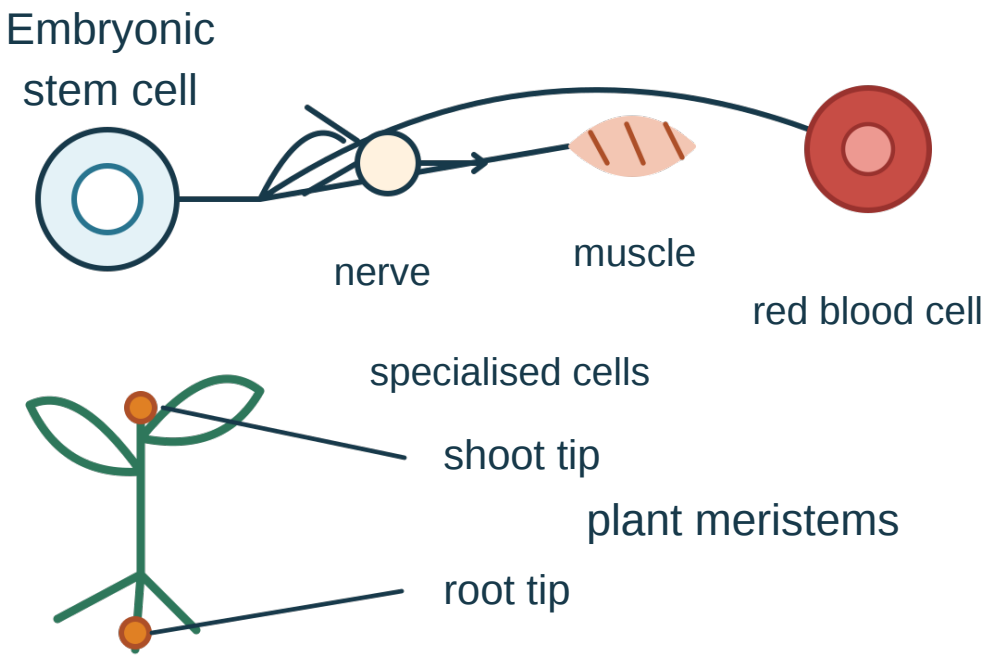
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Arrival · Supported

Look at this week's seedling and skin-repair model. Point to one detail you notice.



An embryonic stem cell can give rise to different specialised cells; meristems at growing plant tips produce new cells. Simplified; not to scale.

Picture colours are illustrative: pale blue stem cell, cream nerve cell, peach muscle cell, red blood cell with a light centre, green plant and orange meristem tips. Colour does not show an observed result.

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Exit ticket · Supported

Show what you know with a model and explain one reason.

Check today’s success:

I can distinguish making more cells from cells taking on a specialised job.

I can compare animal growth with plant growth.

I can identify a role for an animal stem cell or a plant meristem.

What am I learning? I am learning to explain how cells divide and then take on different jobs for growth and repair.

My feedback was … Name what you heard at Review.

Next I need to … Add the step showing how one newly made cell takes on a specialised role.

How I know: I can show division and differentiation and compare the animal and plant growth routes.

Your teacher hears your answer and uses it to choose our next model, prompt or example.

My confidence: □ 1 □ 2 □ 3

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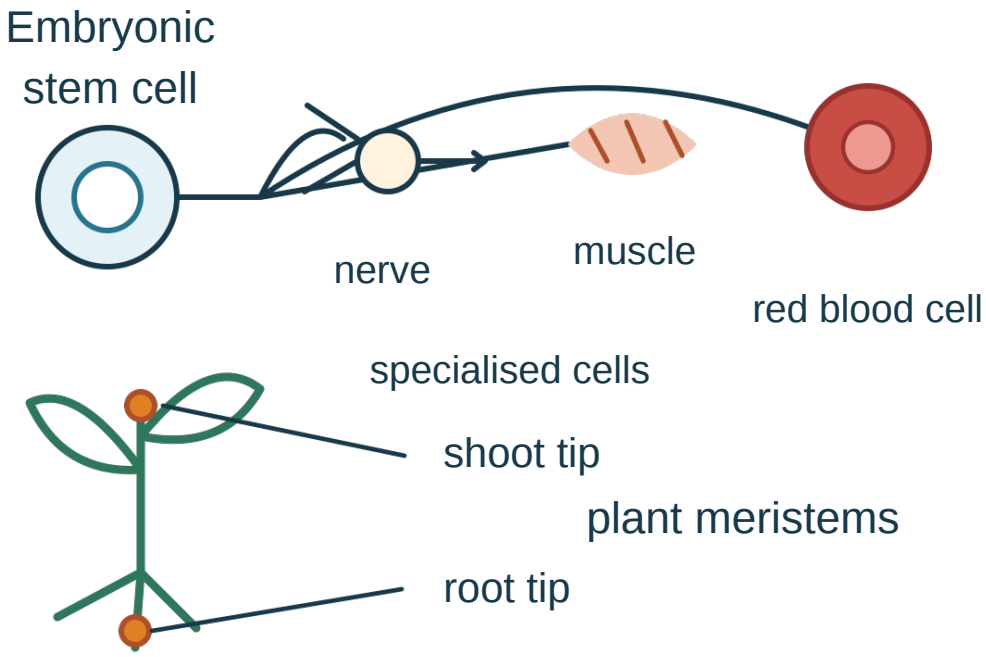
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Exit ticket · Standard

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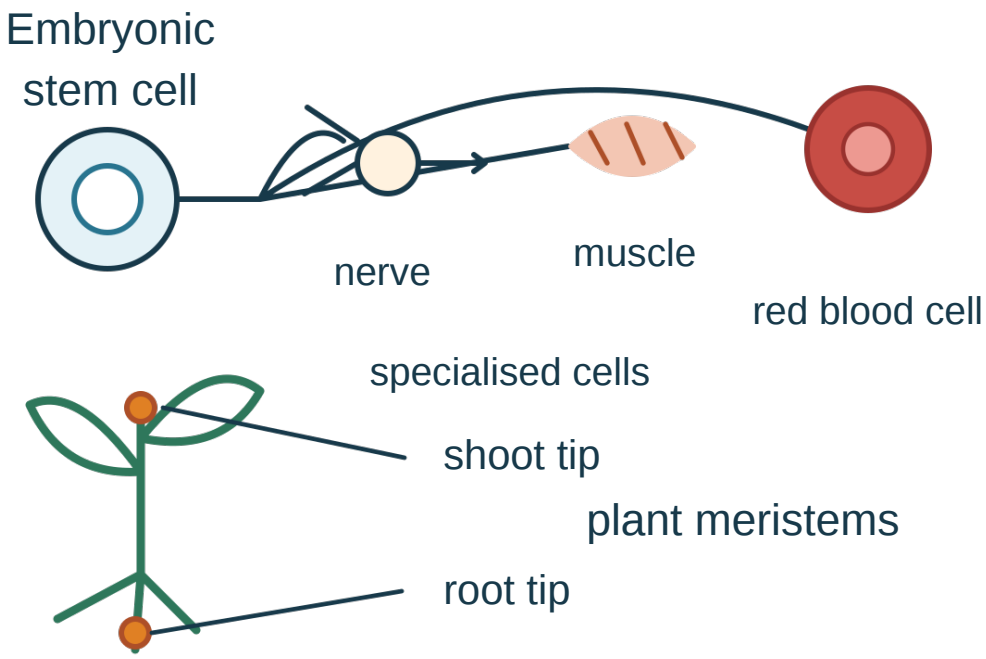
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Arrival · Stretch

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Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

Exit ticket · Stretch

Show what you know with a model and explain one reason.

Check today’s success:

I can distinguish making more cells from cells taking on a specialised job.

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My feedback was … Name what you heard at Review.

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Starter

Recall three ideas from earlier lessons. Show one answer at a time.

What two matching products follow body-cell mitosis and cell splitting?

Which process moves particles from low to high concentration using energy?

In which part of a eukaryotic animal cell are chromosomes found?

Today I can:

I can distinguish making more cells from cells taking on a specialised job.

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I do · words

Watch the full model. Point to the step that answers today's question.

Word	Meaning	Example
differentiation	a cell becoming specialised for a particular job	A developing skin cell differentiates to do a job in skin.
stem cell	a cell that can divide and form specialised cell types	Some adult stem cells make replacement blood cells.
meristem	a plant region whose cells divide to make new cells	A root tip contains a meristem.
growth	an increase in size, including by making more cells	A shoot grows as its cells divide and elongate.

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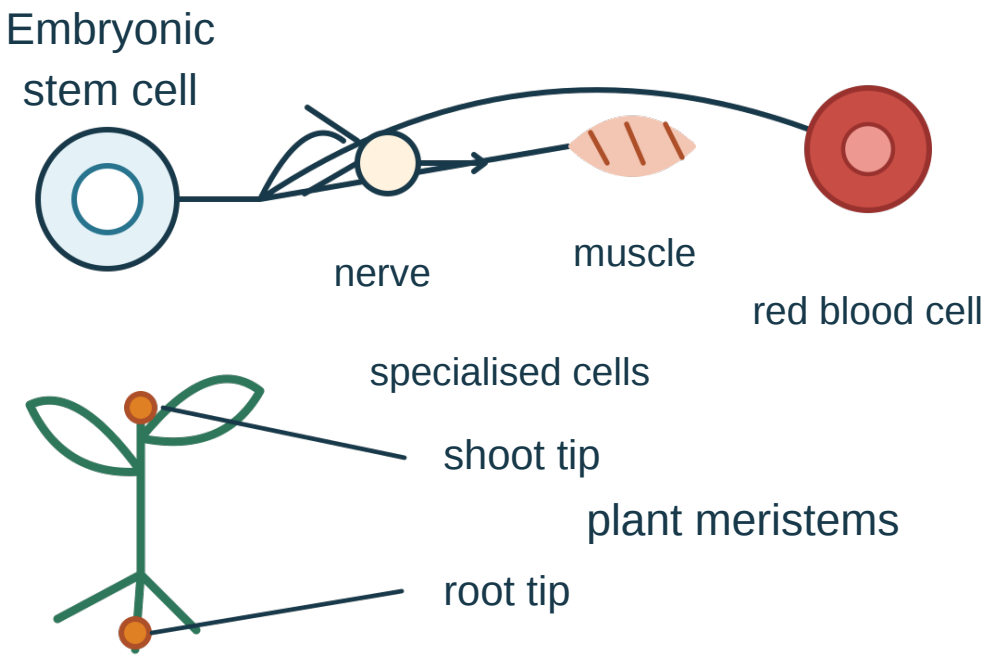
I do · full model

Full model

In skin, mitosis adds cells; some stem-cell descendants differentiate for a tissue job. At a shoot tip, meristem cells divide, then elongate and differentiate.

Show me

Show me: does making more cells on its own mean those cells now have a specialised job?



An embryonic stem cell can give rise to different specialised cells; meristems at growing plant tips produce new cells. Simplified; not to scale.

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We do · missing step and matches

Complete the missing step in the part-done model. Speak, then match the cards.

One-minute reset: use the agreed move, then return to one step.

Part-done model

Skin: mitosis adds cells → differentiation gives a job. Shoot: meristem adds cells → _____ lengthens cells → differentiation.

Speaking: “First cells divide. Then some cells …”

Branch clue	Label choices
Makes more cells	differentiation / division / elongation
Plant cells become longer	differentiation / division / elongation
Cell gains a tissue job	differentiation / division / elongation

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We do · routes

F · support

Use two labelled branches (animal tissue and plant tip); pupils place the missing elongation card and point to the corresponding change.

M · explain

Explain which animal and plant steps both make more cells and which give new jobs.

H · stretch

Explain why a shoot can increase in length even after the division step has ended.

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We do 2

Choose one hinge answer. Both of you show it at the same time.

A gut-lining cell and a nerve cell share inherited DNA but have different structures. Which explanation fits this new pair?

A. They differentiated into specialised cells with different jobs.

B. One cell must have lost every inherited gene.

C. Both cells must perform exactly the same job.

D. Neither cell can have developed through cell division.

Both of you show one answer to the adult at the same time. Point or use a card, too. Then explain to your partner.

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Review and improve

Tell a partner what you changed after feedback. Mark a real edit // and read it back when asked ?.

One-minute reset: use the agreed move, then return to one step.

My feedback was … Name the actual point an adult or partner noticed. Make one true self-edit //. Ask “Read this back to me” ? if that helps.

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Method strip

Try it with kit · prepared model

Kit: Large prepared division, elongation and differentiation cards, desk mat.

Place the prepared cards on the two labelled model branches.

Risk: Loose model cards could fall onto the floor.

Safe: Keep cards on the desk mat and pick up any that fall.

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You do · F

A skin model shows division. What gives a new cell its job?

One-minute reset: use the agreed move, then return to one step.

Speaking: “A plant also …, whereas an animal …”

You can plan in any language.

Show it your way: point, sign, speak, draw, type, write, use a scribe or a staff-captured photo.

Task

Place divide, elongate and differentiate cards on the skin/shoot branches; omit elongate on the skin branch.

In the shoot, first cells …; then they …

I · S · C □

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You do · M

A skin model shows division. What gives a new cell its job?

One-minute reset: use the agreed move, then return to one step.

Speaking: “A plant also …, whereas an animal …”

You can plan in any language.

Show it your way: point, sign, speak, draw, type, write, use a scribe or a staff-captured photo.

Task

Explain why mitosis alone does not give a new cell its specialised tissue job.

More cells are made by …, but a job develops when …

I · S · C □

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You do · H

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One-minute reset: use the agreed move, then return to one step.

Speaking: “A plant also …, whereas an animal …”

You can plan in any language.

Show it your way: point, sign, speak, draw, type, write, use a scribe or a staff-captured photo.

Task

Compare growth in animal tissue with a plant shoot, including an appropriate stem-cell region and what each process adds.

Both organisms …; however, the plant also … because …

I · S · C □

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You do · Go deeper

A skin model shows division. What gives a new cell its job?

One-minute reset: use the agreed move, then return to one step.

Speaking: “A plant also …, whereas an animal …”

You can plan in any language.

Show it your way: point, sign, speak, draw, type, write, use a scribe or a staff-captured photo.

Task

A tissue gains more cells but no new cell types. What does the model explain, and what is missing?

Repair model audit

Correct a diagram labelling new skin cells “stem cell”; justify two changes.

I · S · C □

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Repair model audit · model to check

These supplied labels are a claim to check.

division	→	new skin cells label: stem cell

Correct a diagram labelling new skin cells “stem cell”; justify two changes.

I · S · C □

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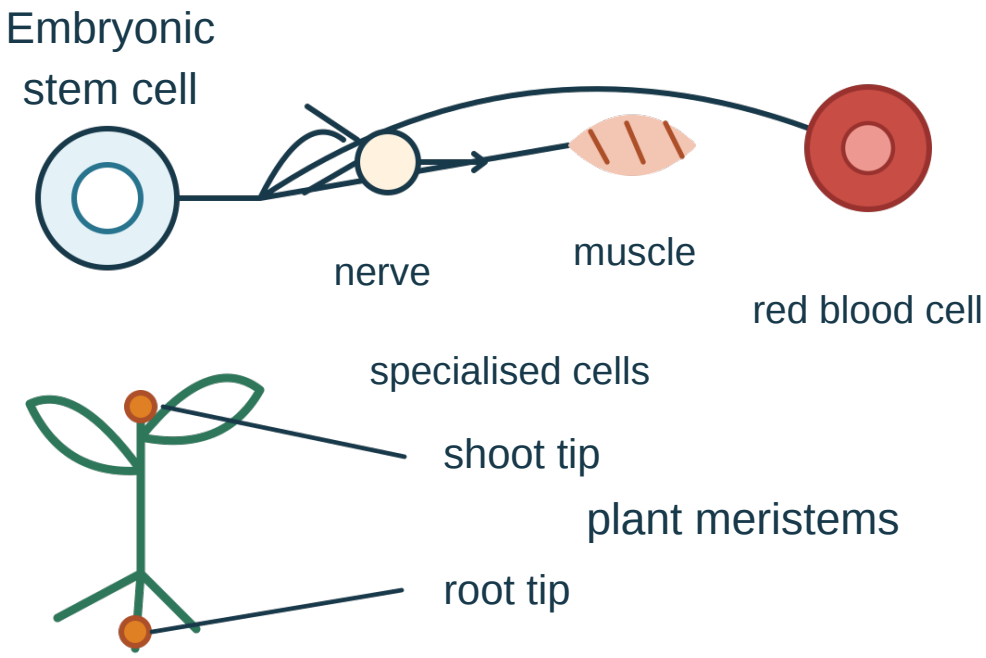
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Knowledge organiser

I am learning to explain how cells divide and then take on different jobs for growth and repair.

Growth and cell jobs

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Knowledge organiser · words and comparison

Word	Meaning
differentiation	a cell becoming specialised for a particular job
stem cell	a cell that can divide and form specialised cell types
meristem	a plant region whose cells divide to make new cells
growth	an increase in size, including by making more cells

Compare the two routes

Skin: mitosis adds cells → differentiation gives a job.

Shoot: meristem adds cells → elongation lengthens cells → differentiation.

Picture colours are illustrative: pale blue stem cell, cream nerve cell, peach muscle cell, red blood cell with a light centre, green plant and orange meristem tips. Colour does not show an observed result.

My language: _____

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Word page

Word	Meaning	Example
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growth	an increase in size, including by making more cells	A shoot grows as its cells divide and elongate.

My language: _____

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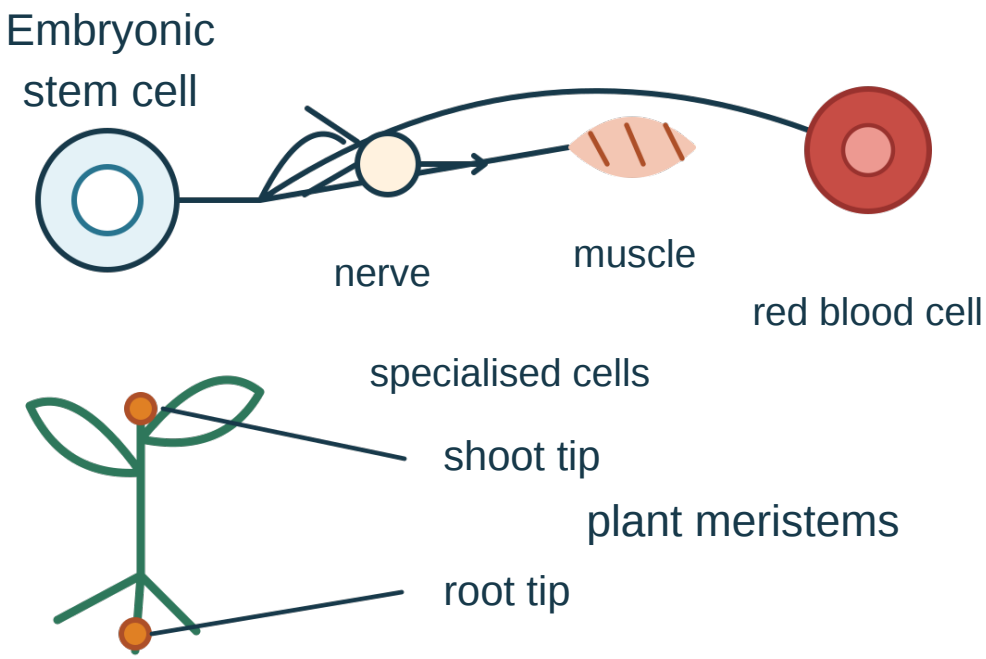
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Picture model



An embryonic stem cell can give rise to different specialised cells; meristems at growing plant tips produce new cells. Simplified; not to scale.

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Cards · match and explain

Clue cards
Makes more cells
Plant cells become longer
Cell gains a tissue job

Label cards
differentiation
division
elongation

Branch labels: animal tissue / plant tip.

Use the cards to build the two routes. Keep each clue and label separate until you choose a match.

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